

# Executive Business Report

## Executive Summary

This report analyzes four years of retail sales data to understand sales trends, identify busy and slow periods, predict sales for the next three months, and recommend better inventory planning. It also highlights unusual sales patterns and groups products based on demand to support smarter stocking decisions. These insights can help reduce stock shortages, avoid excess inventory, and improve overall business performance.

## Three-Month Sales Forecast

Using the XGBoost forecasting model, sales were predicted for the next three months.

| Forecast Month | Predicted Sales |
|----------------|-----------------|
| Month 1        | 54,171          |
| Month 2        | 32,850          |
| Month 3        | 60,098          |

The forecasting models also generated confidence ranges, indicating that actual sales may vary slightly due to market conditions, customer demand, promotional campaigns, or external economic factors. Nevertheless, the forecasts suggest that sales are expected to remain stable, supporting proactive inventory and supply chain planning.

## Top Three Anomalies and Likely Business Causes

The most significant anomalies are likely explained by the following business events:

### 1. Large Sales Spike (November)

Most likely caused by festive shopping, Black Friday promotions, and year-end discount campaigns.

### 2. High Sales During December

Increased customer purchases associated with holiday shopping and corporate year-end procurement.

### 3. Sharp Sales Drop After Peak Season

Likely reflects reduced customer demand following holiday sales, along with inventory normalization after major promotional periods.

These anomalies provide valuable insights for inventory planning and future promotional scheduling.

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## Product Demand Segmentation and Stocking Strategy

| Demand Segment               | Recommended Strategy                                                                  |
|------------------------------|---------------------------------------------------------------------------------------|
| High Volume / Stable Demand  | Maintain higher inventory levels and automate replenishment to avoid stockouts.       |
| Growing Demand               | Gradually increase inventory while monitoring future demand trends.                   |
| Low Volume / High Volatility | Keep lean inventory and replenish only when demand increases to reduce holding costs. |
| Declining Demand             | Reduce stock levels and use promotional discounts to clear existing inventory.        |

These strategies can improve inventory turnover while minimizing storage costs and lost sales.

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## Three Data-Backed Business Recommendations

### Recommendation 1: Use XGBoost as the Production Forecasting Model

XGBoost achieved the best forecasting performance with **MAE = 648.03**, **RMSE = 830.80**, and **MAPE = 1.64%**, significantly outperforming SARIMA and Prophet. Adopting this model will improve forecast accuracy and inventory planning.

### Recommendation 2: Increase Inventory Before Peak Seasonal Months

Historical analysis shows that **November** consistently experiences the highest sales due to seasonal demand. Increasing inventory before this period can reduce stock shortages and maximize sales opportunities.

### Recommendation 3: Prioritize Office Supplies Inventory

Forecasting indicates that **Office Supplies** has the highest projected demand among the analyzed segments. Allocating additional inventory and marketing resources to this category is likely to improve revenue while ensuring product availability.

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## System Limitation / Risk

Although the forecasting model performed very well on historical data, future sales can still be affected by factors that are not included in the dataset, such as competitor actions, economic conditions, pricing changes, marketing campaigns, supply chain disruptions, or unexpected customer behavior.

To maintain forecasting accuracy, the model should be retrained regularly using newly collected sales data and monitored for performance over time.

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## Conclusion

This project successfully developed an end-to-end sales forecasting and demand intelligence system capable of predicting future sales, detecting unusual sales behavior, segmenting products by demand characteristics, and presenting business insights through an interactive dashboard.